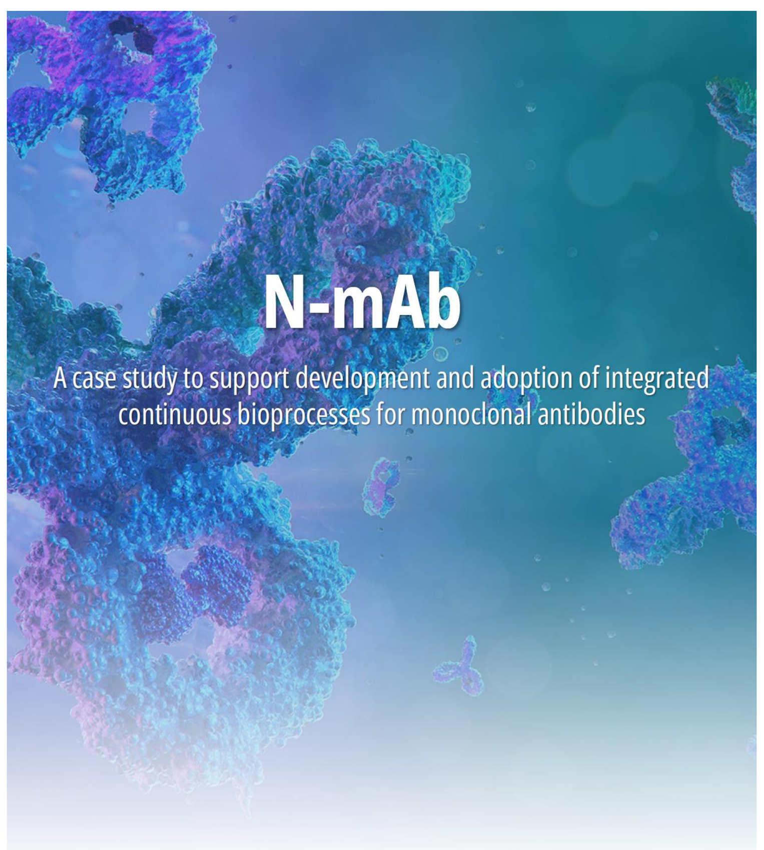

N-mAb: A CASE STUDY SUPPORTING ADOPTION OF INTEGRATED CONTINUOUS BIOPROCESSES

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The A-mAb case study (2009) was created to stimulate discussion around how the core principles of QbD could be applied to the process development of a monoclonal antibody with examples of a multitude of real-world scenarios, as opposed to a singular approach. A-mAb was very successful in generating that discussion and impacted how process development was executed across the industry. Following a workshop on Process Intensification held at the National Institute for Innovation in Manufacturing Biopharmaceuticals (NIIMBL) in early 2020, it was decided to create a case study to support teaching and learning for both industry and regulators around adoption of advanced manufacturing process technologies such as integrated continuous bioprocesses for mAbs. This case study, named N-mAb, would provide examples of the implementation of different process options and associated process technologies that can drive an integrated control strategy, including considerations related to process development and characterization, process validation, and deviation management consistent with current guidance in place. Like its predecessor, A-mAb, the goal is to stimulate discussion and advance new concepts around integrated continuous bioprocesses. To make the discussion in this case study more manageable and still retain some diversity in process design, two major process options are discussed, based on the presence or absence of an intermediate batch surge tank as shown below, along with several minor options. To illustrate different scenarios and the associated decisions involving variation in product quality, examples are developed based on an informed mock dataset which was generated from consensus industry experiences. N-mAb follows the evolution of an integrated control strategy from early clinical through process validation and commercial manufacturing as process understanding increases. It is important to note that the evolving versions of the control strategy are essential to assessing deviations and other aspects of managing the process in real time. A unique difference from A-mAb is that this N-mAb case study includes a discussion of the challenges involved in managing quality in real time as necessitated by a continuous process Since the control of adventitious agents often represents a unique subset of the overall control strategy and can be uniquely challenging for a continuous process, it is presented as a separate chapter. Another chapter expands on how decision processes can be constructed to manage deviations in either CPPs or CQAs. Examples of how the response to deviations can inform updates to a control strategy are given to demonstrate how continued learning cycles build upon the process knowledge established during Stage 1 Process Characterization. Overviews of these chapters as well as the threads linking the stages in the development of an integrated control strategy will be presented in the poster.

KEY TERMINOLOGY: BATCH

It is possible to leverage the following definition of a batch from ICH Q7:

"A specific quantity of material produced in a process or series of processes so that it is expected to be homogeneous within specified limits. In the case of continuous production, a batch may correspond to a defined fraction of the production. The batch size can be defined either by a fixed quantity or by the amount produced in a fixed time interval."

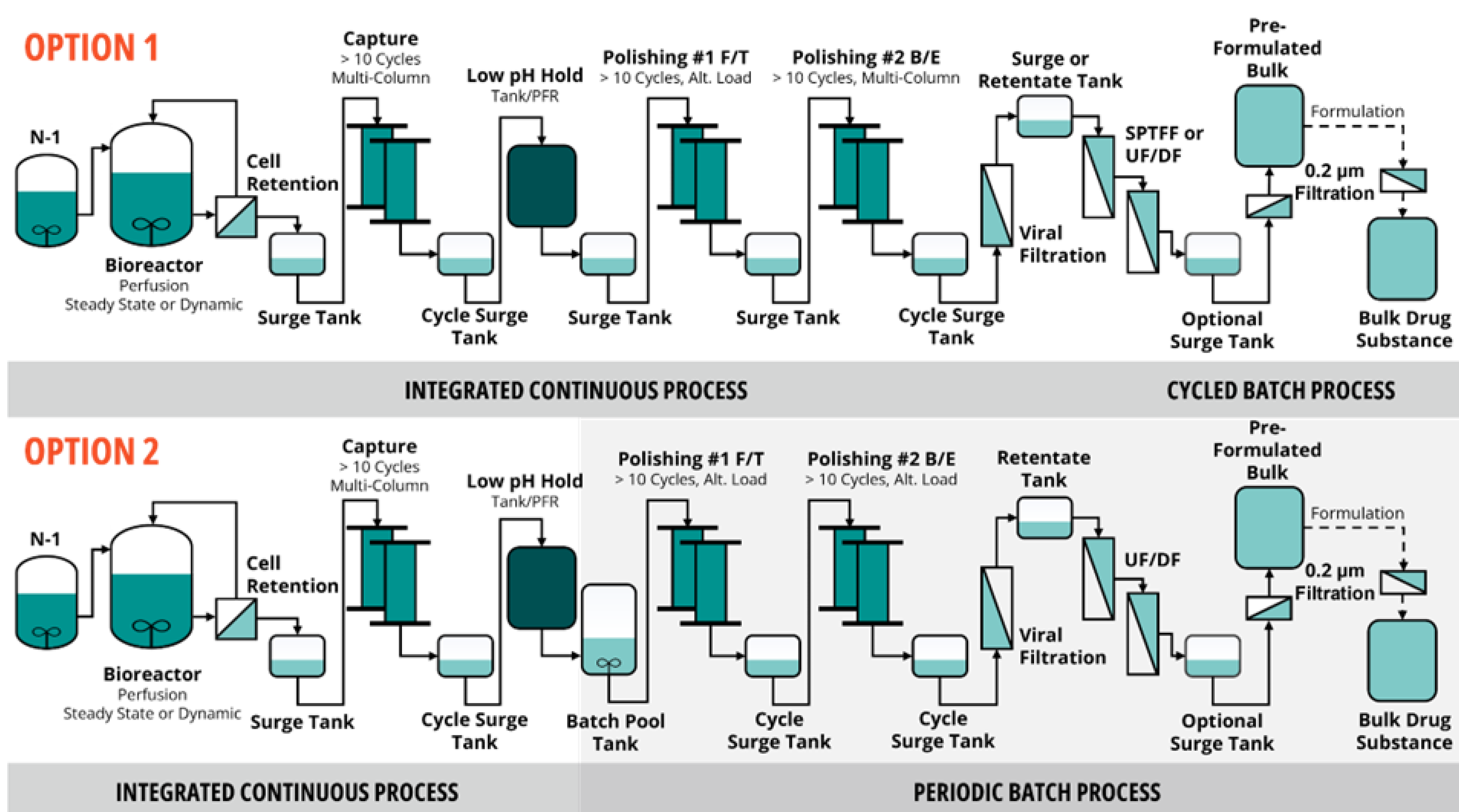
However, for this N-mAb document, an operational definition of a drug substance batch which is slightly more detailed is preferred:

"A specific quantity of purified product of interest having a unique identifier that enables traceability of raw materials, production bioreactor days, subsequent downstream unit operations, and a pre-defined, unique data set confirming adherence to in-process control limits and final product quality release specifications that enable disposition by quality systems and forward processing to drug product."

Key Conclusions:

- A batch is defined by a specific amount of material AND
- A pre-specified package of information related to the integrated control strategy that allows for assurance of product quality

Note that the definition of a batch is not necessarily linked to the harvest duration of a single bioreactor. A single bioreactor could generate multiple downstream batches, or harvest streams from multiple bioreactors could be combined into one or more downstream batches.



Two major options for integrated continuous bioprocesses discussed in N-mAb

KEY TERMINOLOGY: SURGE TANK

SURGE TANK

Purpose:

- Serve as break tank
- Accommodate small fluctuations in flow between unit ops
- Manage pressure differences

Residence Time: Minutes to tens of minutes

Key Consideration: May lead to material loss in case of significant disruption at unit op interface

CYCLE SURGE TANK

Purpose:

- Accumulate one (or a few) sub-cycles exiting a unit operation.
  - Useful for unit op with highly variable output, e.g., bind and elute
- Allows for homogenization before passing to next unit op
- Designed for cyclic output volume from a unit op

Residence Time: Hours

Key Consideration: May impact residence time broadening, traceability

BATCH POOL TANK

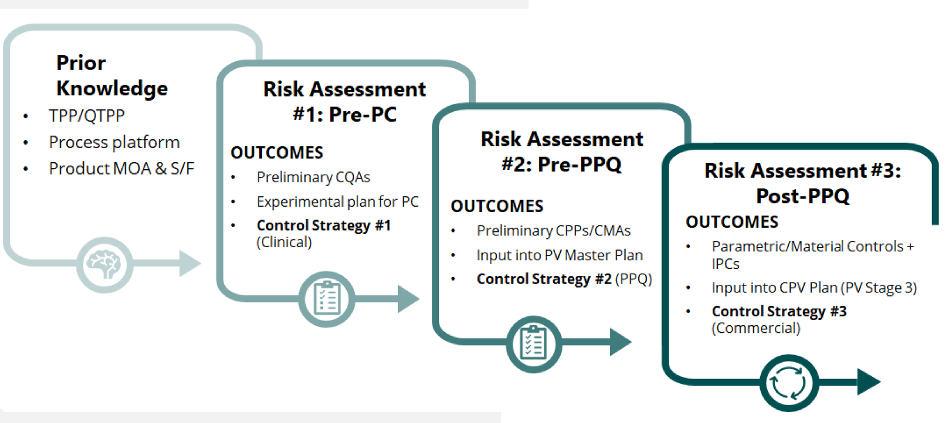
Purpose:

- Accumulate material equivalent to the mass of a single drug substance lot
- Allows for collection and homogenization
- Minimize impact of upstream product quality variability

Residence Time: 24hr or more

Key Considerations:

- Typical locations include post-Protein A or after UF/DF
- May require a large vessel, demonstration of stability



Process Design Elements Covered in N-mAb:

1. **Understanding & analysis of quality & performance attribute dynamics**
2. **Understanding of physical dynamics of system**
3. **Implementation and verification of adventitious ant control strategy**
4. **Verification of control using appropriate scale runs**
5. **Strategy for managing the process in real time (RT decisions)**
6. **Learning cycle for product & process understanding**
7. **Integrated Control Strategy for Commercial Manufacturing**

Process Design Drives

Integrated Control Strategy

Evolution of Control Strategy over Process Lifetime

Control Strategy #1 Initial Clinical Supplies

Typically based on platform process

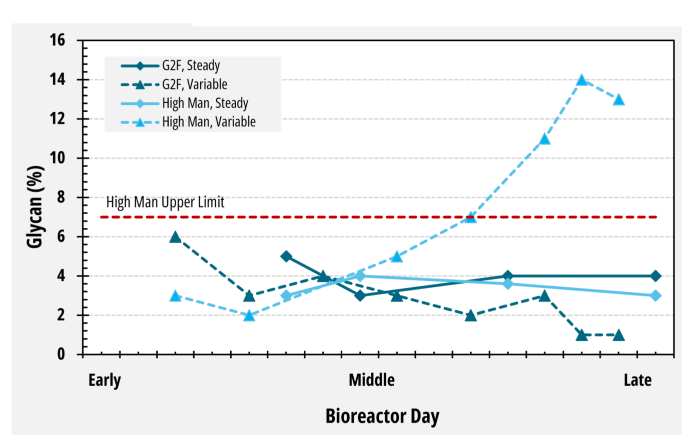
Control Strategy #2 Prior to Registrational Batches/PPQ

Updated based on PC studies and any at or near scale experience

Control Strategy #3 Commercial Manufacturing

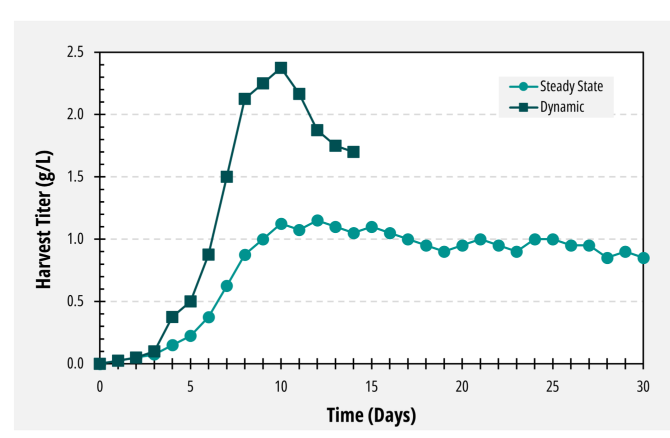
Updated based on PPQ and responses to HA questions

1



Variation in product glycosylation patterns for G2F and high mannose in cultures with steady (solid lines) and variable (dashed lines) product quality

1

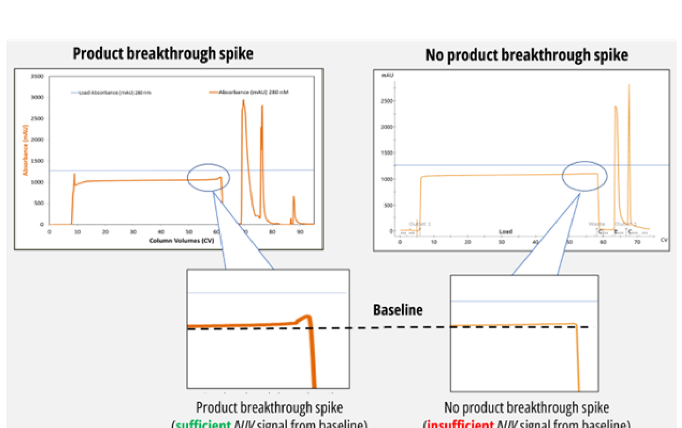


Product titer profile example for dynamic and steady-state perfusion

1

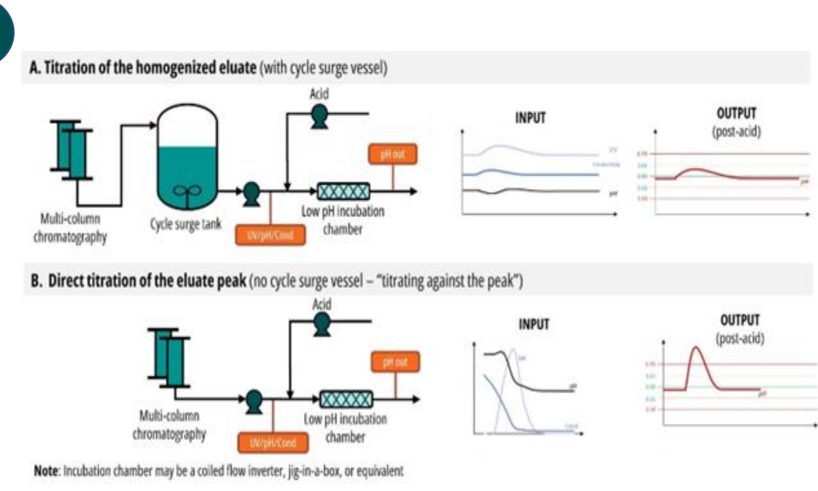
Method	Considerations and Approach
Time Dependency Assessment	• Assess input parameters for time dependency • Apply univariate control charts • Evaluate to assess time dependency (i.e., ANOVA of time as a significant effect on parameter or set variance criteria)
Standard DOE/RSM Model approach for non-time-dependent parameters	• Apply to non-time-dependent parameters • Define criteria for acceptable output parameter variation based on downstream capabilities • Select approximately 3–5 days during the production phase to generate individual DOE/RSM prediction models and compare outcomes across timepoints • Perform ANOVA evaluation of time as a significant effect on outcomes across days
Mixed Model Repeated Measurement for time-dependent parameters	• Apply to time-dependent parameters or use to further justify the lack of time dependence • Apply to fixed parameter OFAT or DOE studies • Apply covariance of time factor to model
Bayesian Hierarchical Model	• Model linear and non-linear relationships • Estimate time along with other control parameters as random variables • Evaluate posterior distribution of time and fixed control parameters for significance • Estimate probability of specific outcomes using posterior predictive distribution

2



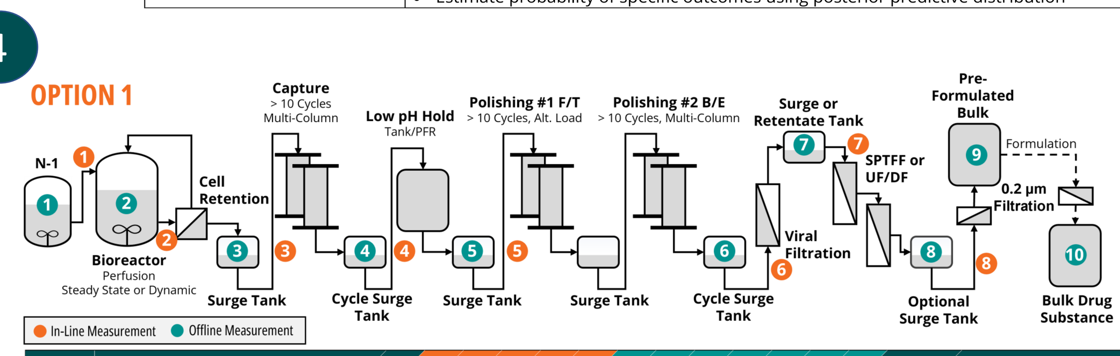
Examples of breakthrough curves leveraged for deltaUV-based load control. The orange line represents outlet UV, the blue line represents load UV, and the black dashed line indicates the load breakthrough baseline.

2



Comparison of simulated input variability and output control of pH for virus inactivation with and without cycle homogenization (titrating against the peak).

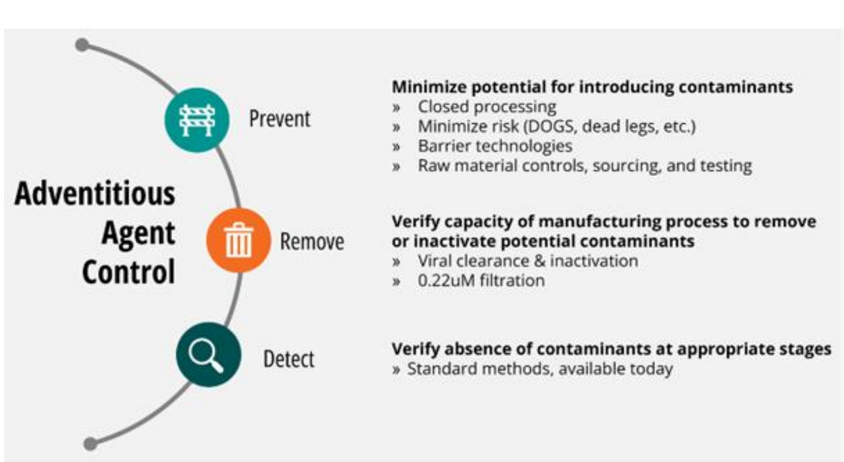
4



Sample Point	Process Step	Elapsed Time (h)	Capacitance	Inline A280	Fluid Mass Flow	pH	Conductivity	VCD/Viability	Metabolites	Titer/A280	Glycosylation	Deamidation	HMWS	HCP	Initial Risk (of variance)	Sampling Frequency: Initial Runs
			Performance Attribute							Quality Attribute						
1	N-1		•					•							Low	Last day
2	Bioreactor		•					•	•						High	Every 2d after inoculation
			•							•	•	•	•			Every 4d after start of perfusion
3	Post-ATF/TFF Permeate Surge Tank	0			•					•					Low	1. Prior to 1st capture cycle 2. Once every 24h 3. Prior to last capture cycle
4	Post-ProA Surge Tank	8		•	•	•	•			•		•	•		Medium	1. Prior to 1st capture cycle 2. Twice weekly, after cycle homogenization 3. Prior to last capture cycle
5	Post-VI Surge Tank	12		•	•	•	•			•		•	•		Low	1. After 1st cycle 2. After last cycle
6	Post-Polishing Chrom Pre-VF Surge Tank	14		•	•	•	•			•		•	•		Low	1. After 1st cycle 2. Sample from the surge tank corresponding to each lot of planned DS 3. After last cycle
7	Post-VF, Pre-UFDF Surge	16		•	•	•	•			•					Low	Every 24h
8	Post-UFDF, Pre-0.2 µm	18		•	•	•	•			•					Low	Every 24 h
9	Post-UFDF, Post-0.2 µm filter	18								•					Low	Representative sample of every bag filled
10	Bulk DS									•	•	•	•	•	Low	Representative sample of every bag filled

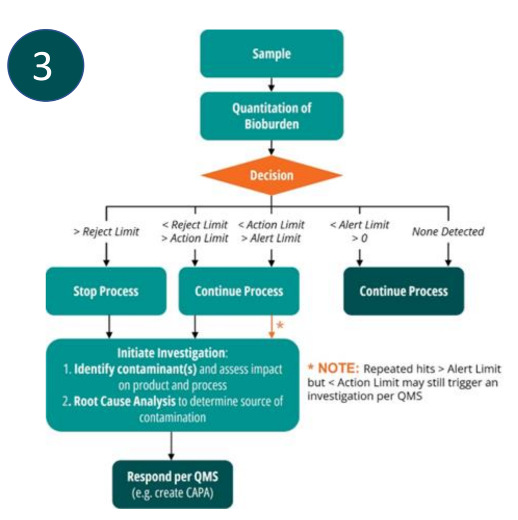
Analytical sampling and testing plan for initial clinical runs through PPQ.

3



A complementary approach ensures microbial and viral safety of biological products

3



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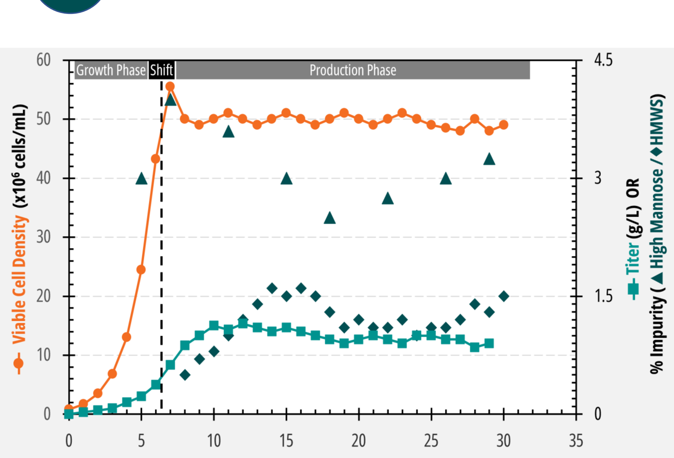
CQA	Potential Source, Stress Response and Clearance	Process Controls: Parametric Controls & Material Attributes	Process Capability	Analytical Controls: Testing Strategy including IPCs	Residual Risk
Glycosylation: Galactosylation	Source: Bioreactor Stress Response: None Stability Indicating: No Clearance: None	Bioreactor PPs: Temp, pH, DO, harvest day Bioreactor MAs: Trace metals (Cu & Mn)	High	DS Release Testing	Low
Glycosylation: Fucosylation	Source: Bioreactor Stress Response: None Stability Indicating: No Clearance: None	Bioreactor PPs: Temp, pH, DO, harvest day Bioreactor MAs: Trace metals (Cu & Mn)	Medium	DS Release Testing	Low
Glycosylation: High Mannose	Source: Bioreactor Stress Response: None Stability Indicating: No Clearance: None	Bioreactor PPs: Temp, pH, DO, harvest day Bioreactor MAs: Trace metals (Cu & Mn)	High	DS Release Testing	Low
Deamidation at Asn325	Source: Bioreactor, DSP Stress Response: pH, heat Stability Indicating: Yes Clearance: None	Bioreactor PPs: Temp, pH, DO, harvest day DSP PPs: DS/Intermediate hold time & temp	High	Discontinue Release and Stability Testing – Deamidation at ASN325 was determined to be not clinically relevant	Low
High Molecular Weight Species (HMWS)	Source: Bioreactor, DSP Stress Response: pH, heat, shaking, light, metals, freeze/thaw Stability Indicating: Yes Clearance: CEX = 3-fold	Bioreactor PPs: Temp, pH, DO, harvest day DSP PPs: VI: pH & time CEX: protein load DSP CMAs: CEX elution buffer pH Trace metal leaching Procedural Controls: Protect process intermediates from light when risk of exposure is high; Minimize vortexing and foaming in surge tanks	High	Consider control of CEX load based on IPC for HMWS post-VI to maximize yield DS Release Testing DS Stability Testing	Low
Host Cell Protein (HCP)	Source: Bioreactor Stress Response: None Stability Indicating: No Clearance: Chrom Step(s)	Bioreactor PPs: Temp, pH, harvest day DSP PPs: protein load, peak cutting	High	DS Release Testing	Low

4



Example data from initial clinical run at full scale.

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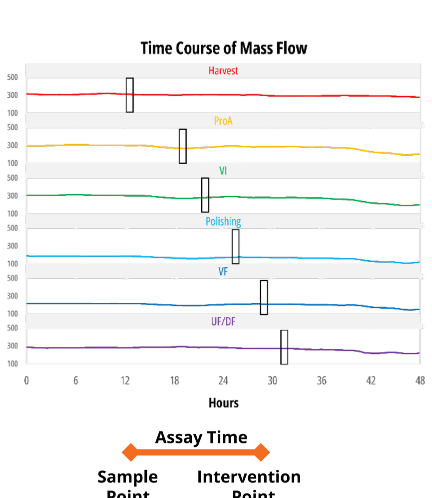

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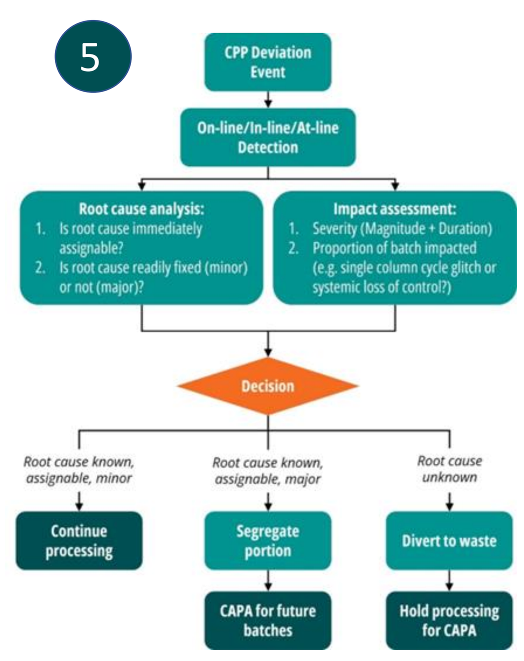

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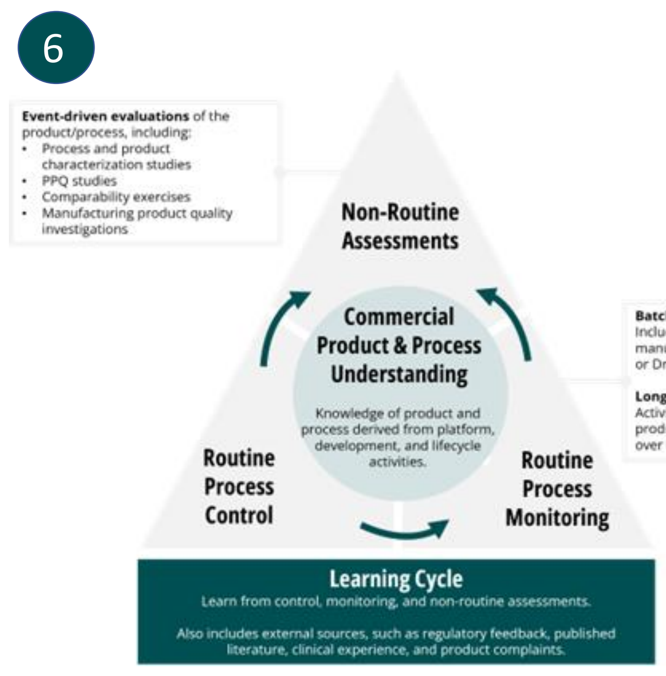

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